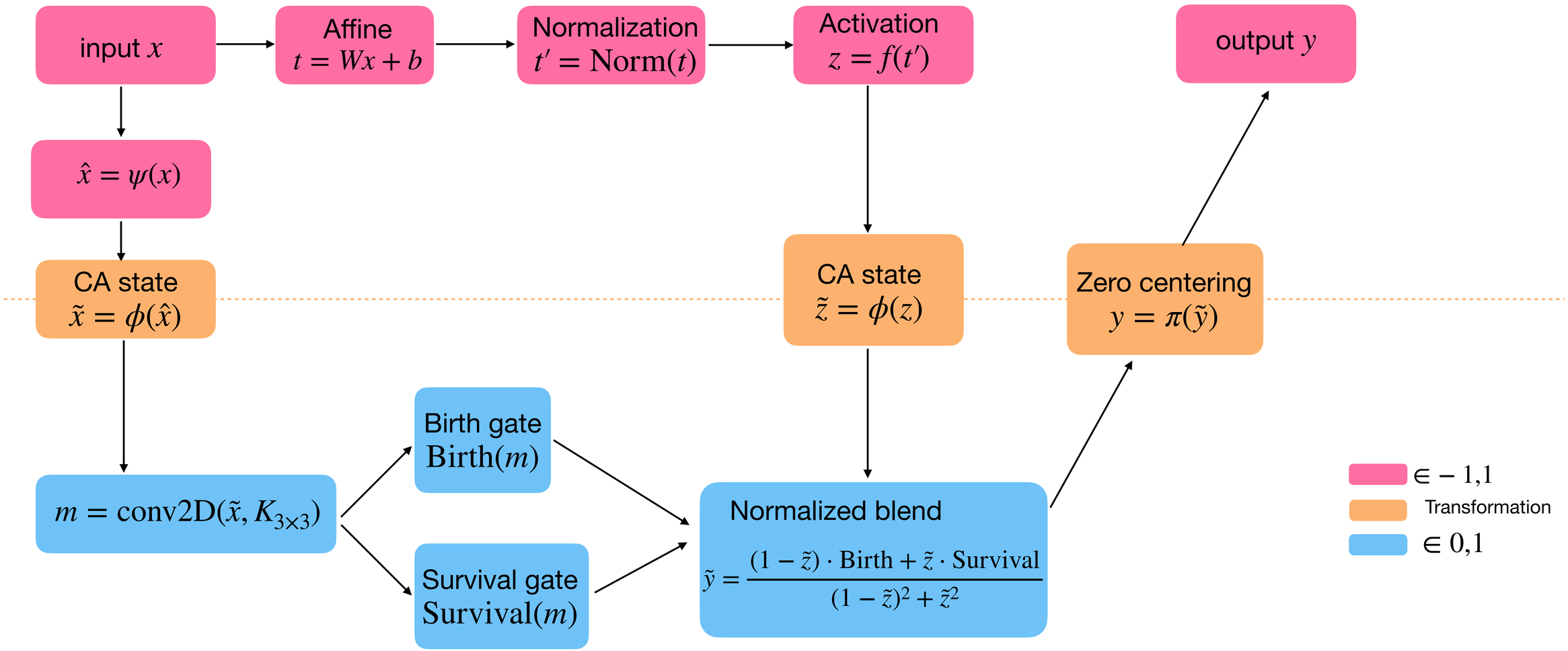

input x
Affine
$t = Wx + b$
Normalization
$t' = \mathrm{Norm}(t)$
Activation
$z = f(t')$
output y
$\hat{x} = \psi(x)$
CA state
$\tilde{x} = \phi(\hat{x})$
CA state
$\tilde{z} = \phi(z)$
Zero centering
$y = \pi(\tilde{y})$
Birth gate
$\mathrm{Birth}(m)$
$m = \mathrm{conv2D}(\tilde{x}, K_{3\times3})$
Survival gate
$\mathrm{Survival}(m)$
Normalized blend
$\tilde{y} = \frac{(1-\tilde{z})\cdot \mathrm{Birth} + \tilde{z}\cdot \mathrm{Survival}}{(1-\tilde{z})^2 + \tilde{z}^2}$
$\in -1,1$
Transformation
$\in 0,1$